

Do Biomedical Tasks Require Biomedical Foundation Models?

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Abstract

Foundation models are increasingly used across biomedical domains, including clinical text, medical imaging, genomics, and protein modeling. At the same time, rapidly advancing general-purpose foundation models can often be adapted to biomedical tasks through prompting, fine-tuning, and tool use. This raises a central question: **do biomedical applications require domain-specific foundation models, or can general models be adapted effectively?**

In this review, we examine the landscape of biomedical foundation models and compare domain-specific pretraining with adaptation of general models. We discuss how these approaches interact with biomedical data systems and the challenges of evaluation, reliability, and deployment. We conclude by outlining key open problems that will shape the future of foundation models in biomedical research and healthcare.

Introduction

The Current State of Domain-Specific Biomedical Foundation Models

Overview of foundation models trained directly on biomedical data across major modalities such as clinical text, imaging, genomics, proteins, and EHRs.

Adapting General Foundation Models to Biomedical Tasks

How general-purpose foundation models (e.g., large language and vision models) are adapted to biomedical applications through prompting, fine-tuning, and tool use.

Integrating Foundation Models with Biomedical Data Systems

How foundation models interact with structured biomedical data and knowledge sources, including electronic health records, ontologies, and databases.

Evaluation, Reliability, and Deployment of Biomedical Foundation Models

Challenges in deploying and evaluating biomedical foundation models and ensuring reliability, including benchmarking, dataset bias, interpretability, and clinical validation.

Open Problems and Future Directions

Despite substantial progress in biomedical foundation models, several fundamental challenges remain unresolved. Building on the central themes of this review—namely the tension between domain-specific and general models, the importance of system-level integration, and the need for reliable evaluation—we highlight three core open problems that are likely to shape the future of the field. These challenges are deeply interconnected and reflect limitations not only in model design, but also in data, training objectives, and deployment environments.

Domain-Specific Modeling versus General Adaptation

A central unresolved question is whether biomedical tasks inherently require domain-specific foundation models, or whether increasingly capable general-purpose models can be adapted effectively. Domain-specific models, such as BioBERT and ClinicalBERT [1,2], as well as highly specialized systems like AlphaFold [3], demonstrate that incorporating domain knowledge into pretraining can yield substantial gains, particularly when the underlying structure of the problem is well understood. These successes suggest that inductive biases tailored to biomedical data remain highly valuable.

However, recent advances in large-scale general foundation models have challenged this view. Models such as GPT-4 and PaLM have shown promising performance on some biomedical reasoning tasks with minimal domain-specific training [4,5,6], raising the possibility that sufficiently large models may implicitly learn transferable structure. At the same time, these models often exhibit hallucinations and lack reliable grounding in biomedical knowledge [7], indicating that scale alone may not resolve domain-specific challenges. For instance, a general model might fail to distinguish between clinically important subtypes of a disease that require different treatment strategies.

The absence of a principled framework for resolving this trade-off remains a major limitation. Without such understanding, model development is driven largely by empirical performance on isolated benchmarks rather than by a deeper characterization of task requirements. This motivates a critical direction for future research: identifying the conditions under which domain-specific structure is essential versus when it can be effectively approximated through adaptation. Progress in this area is necessary not only for improving performance, but also for reducing the substantial computational and data costs associated with training specialized models from scratch. In particular, approaches that combine general representations with domain-specific grounding, such as retrieval-augmented systems [8], are promising precisely because they attempt to reconcile these competing paradigms rather than choosing between them.

Misalignment Between Pretraining Objectives and Biomedical Utility

A second major challenge arises from the mismatch between standard pretraining objectives and the requirements of real-world biomedical applications. Most foundation models are trained using self-supervised objectives, such as next-token prediction, which encourage the learning of general-purpose representations from large-scale data [9,10]. While this paradigm has been remarkably successful, it does not directly optimize for the types of decisions that biomedical systems are expected to support, such as diagnosis, risk prediction, or treatment planning. This limitation arises because next-token prediction is designed to model the distribution of observed text rather than to capture decision-relevant or causal structure. As a result, models may generate fluent and contextually plausible outputs without ensuring correctness or reliability in high-stakes biomedical settings.

This misalignment becomes evident during deployment. Models that perform well on standard benchmarks may still produce outputs that are clinically unsafe, poorly calibrated, or difficult to interpret [7,11]. As a result, extensive post-training procedures—including supervised fine-tuning, instruction tuning, and reinforcement learning from human feedback—are required to align model behavior with user expectations [12,13]. However, these approaches rely heavily on curated datasets and human supervision, which are both costly and difficult to scale in biomedical contexts.

The persistence of this gap suggests that current training paradigms are fundamentally incomplete. Future progress therefore depends on developing methods that more directly align model training with downstream utility. This includes not only improved alignment techniques, but also a rethinking

of objective functions to incorporate domain-specific notions of correctness, uncertainty, and risk. Such developments are essential because biomedical applications are inherently high-stakes, and models that optimize proxy objectives without capturing real-world utility may fail in ways that are both subtle and consequential. For example, predicting ICD codes is not equivalent to performing diagnosis. ICD codes are primarily designed for administrative and billing purposes and may only imperfectly reflect the underlying clinical reasoning process. This highlights the gap between optimizing models for proxy labels and developing systems that support clinically meaningful decisions, which in turn raises challenges for interpretability and clinical trust.

Robustness Under Distribution Shift and System-Level Constraints

A third critical challenge is the lack of robustness of foundation models under distribution shift, particularly in the complex environments characteristic of biomedical data. Unlike many standard machine learning benchmarks, biomedical datasets are generated by heterogeneous and evolving processes. Clinical data, for example, reflects a combination of biological variation, patient behavior, clinician decision-making, and institutional practices [14,15]. As a result, models trained on one dataset often fail to generalize reliably to others. For example, models trained on data from one hospital may perform poorly when applied to another due to differences in patient populations, clinical practices, or data collection protocols, such as variation across imaging scanners or electronic health record systems.

This problem is not merely statistical but fundamentally structural. The underlying data-generating processes are often unstable and context-dependent, meaning that standard assumptions of independent and identically distributed data do not hold. In addition, deployment environments impose constraints that are rarely captured during model development, including integration with electronic health records, regulatory requirements, and the need for interpretability and user trust. In particular, interpretability plays a central role in biomedical settings, where model outputs must be understood and justified for clinical decision-making, error analysis, and regulatory approval.

These considerations suggest that improving model performance in isolation is insufficient. Instead, future progress requires a shift toward system-level thinking, in which models are designed as components within larger biomedical infrastructures. An important question at this level is whether biomedical AI should rely on a single unified multimodal model, or on modular systems that combine modality-specific models through a downstream decision component. While unified models may enable richer cross-modal representations, modular approaches may offer advantages in debugging and clinical integration. The relative merits of these approaches remain unclear, and identifying when each paradigm is preferable is an important open question.

This architectural perspective naturally motivates research into methods for handling distribution shift, such as domain adaptation and out-of-distribution detection [16], as well as approaches that incorporate causal structure or external knowledge to improve robustness. The importance of these directions stems from the fact that reliability in biomedical AI depends not only on predictive accuracy, but on the ability of systems to function consistently across diverse and changing real-world settings.

Summary

Taken together, these challenges highlight the need for a more integrated understanding of biomedical foundation models. The key open problems identified here—balancing domain specificity and generality, aligning training objectives with real-world utility, and ensuring robustness under distributional and system-level constraints—are not independent issues, but rather reflect different aspects of the same underlying tension between model capability and practical deployment.

Addressing these challenges will be essential for translating recent advances in foundation models into reliable and impactful biomedical applications.

Conclusions

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